Unless explicitly requested by a problem, do not include sketches as part of your proof. You are free to use the result from any problem on this (or previous) assignment as a part of your solution to a different problem even if you have not solved the former problem.

Problem 1 (0.5 + 0.5 pts)**.** Recall that K_{n_1,n_2,n_3} is the complete tripartite graph with parts of sizes n_1, n_2, n_3.

Fix any positive integer n.

(a) Prove that $K_{n,2n,3n}$ is Hamiltonian.

(b) Prove that $K_{n,2n,3n+1}$ is not Hamiltonian.

Solution. (a) Let A, B and C denote the parts with sizes $|A| = n$, $|B| = 2n$, $|C| = 3n$. Thus, for $a \in A$, $b \in B$ and $c \in C$, we have

$$\deg a = 2n + 3n = 5n, \qquad \deg b = n + 3n = 4n, \qquad \deg c = n + 2n = 3n.$$

As a consequence $\delta(K_{n,2n,3n}) = 3n$. Since $|V(K_{n,2n,3n})| = n + 2n + 3n = 6n$, we certainly have $\delta(K_{n,2n,3n}) \geqslant \frac{1}{2}|V(K_{n,2n,3n})|$. Thus, by Dirac's condition, $K_{n,2n,3n}$ is Hamiltonian.

(b) Let A, B and C denote the parts with sizes $|A| = n$, $|B| = 2n$, $|C| = 3n + 1$. Since C is an independent set, we see that the graph $K_{n,2n,3n+1} - (A \cup B)$ is the graph on vertex-set C with no edges. Thus, the number of components of $K_{n,2n,3n+1} - (A \cup B)$ is $|C| = 3n + 1$. However, $|A \cup B| = n + 2n = 3n < 3n + 1$, thus by Theorem 6.5, $K_{n,2n,3n+1}$ can not be Hamiltonian. □

Problem 2 (1 + 2 pts)**.** Fix any integer $n \geqslant 3$.

(a) Construct an n-vertex graph G with $\binom{n-1}{2} + 1$ many edges such that G is *not* Hamiltonian.
(Note: You must construct such a graph for *every* $n \geqslant 3$.)

(b) Prove that if G is an n-vertex graph with at least $\binom{n-1}{2} + 2$ many edges, then G is Hamiltonian.
(Hint: Ore. Also, a graph on k-vertices has at most $\binom{k}{2}$ edges, and $\binom{n-1}{2} - \binom{n-2}{2} = n - 2$)

Solution. (a) Consider a copy of the clique K_{n-1} on vertices $\{v_1, \ldots, v_{n-1}\}$, and define G by adding a leaf u adjacent only to v_1. Note that G can not be Hamiltonian, because that requires $\delta(G) \geqslant 2$. Also, we see that the number of edges is

$$|E(G)| = 1 + |E(K_{n-1}) = 1 + \binom{n-1}{2}.$$

(b) Since the complete graph K_n for any $n \geqslant 3$ is Hamiltonian, we can just assume that $G \ncong K_n$. Fix nonadjacent vertices u, v, and consider the graph G' obtained from G by deleting the vertices u and v. Since u and v are nonadjacent, we have $|E(G)| = |E(G')| + \deg_G u + \deg_G v$, or in other words,

$$\deg_G u + \deg_G v = |E(G)| - |E(G')|.$$

Moreover, since G' has exactly $n - 2$ vertices, we have $|E(G')| \leqslant \binom{n-2}{2}$. Hence

$$\deg_G u + \deg_G v = |E(G)| - |E(G')| \geqslant \binom{n-1}{2} + 2 - \binom{n-2}{2} = n.$$

Since $n \geqslant 3$, we deduce from Ore's condition that G is Hamiltonian. □

Problem 3 (2pts)**.** Let G be a graph on $n \geqslant 4$ vertices with the property that $N(u) \cup N(v) \supseteq V(G)\backslash\{u, v\}$ for every $u \neq v \in V(G)$. Prove that G is Hamiltonian.

Solution. We show that G satisfies Dirac's condition. Fix any $v \in V(G)$. We claim that $\deg v \geqslant n-2$. If not, then $\deg v \leqslant n-3$, and so $|\{v\} \cup N(v)| \leqslant n-2$. But then we can find two distinct vertices x, y which are *not* in $\{v\} \cup N(v))$. Now, $v \in V(G)\backslash\{x, y\}$, but $x, y \notin N(v)$, which implies that $v \notin N(x) \cup N(y)$, a contradiction.

Thus, we have $\delta(G) \geqslant n-2$, and also $n-2 \geqslant n/2$ because $n \geqslant 4$. Therefore $\delta(G) \geqslant n/2$, i.e. G satisfies Dirac's condition. Since $n \geqslant 3$, G is Hamiltonian. □

Problem 4 (2pts)**.** Prove that $\alpha(G) + \beta(G) = n$ for any n-vertex graph G.

(Note: You don't need to know anything about matchings to prove this.)

Solution. We show first that $\alpha(G) + \beta(G) \leqslant n$. To do so, let $A \subseteq V$ be a maximum independent set in G, so $|A| = \alpha(G)$. Since A is an independent set, no edge of G can have both end-points in A and so every edge of G has at least one end-point in $V\backslash A$. In other words, $V\backslash A$ is a vertex-cover of G, and so $\beta(G) \leqslant |V\backslash A| = n - \alpha(G)$. Thus $\alpha(G) + \beta(G) \leqslant n$.

Now, to see that $\alpha(G) + \beta(G) \geqslant n$, let B denote a minimum vertex-cover of G, so $|B| = \beta(G)$. Since B is a vertex-cover, every edge of G has at least one end-point in B. Therefore, no edge has both end-points in $V\backslash B$, and so $V\backslash B$ is an independent set in G. Therefore, $\alpha(G) \geqslant |V\backslash B| = n - \beta(G)$, thus $\alpha(G) + \beta(G) \geqslant n$. □

Problem 5 (2 pts)**.** Let G be a bipartite graph on n vertices. Show that $\alpha(G) = n/2$ if and only if G has a perfect matching.

Solution. Let U, W denote the parts of G. Since $V(G) = U \sqcup W$, one of these sets has at least $|V(G)|/2$ elements. Since these sets are independent, we have $\alpha(G) \geqslant |V(G)|/2$.

($\Leftarrow$) If G has a perfect matching, then every independent set contains at most one vertex from each of the edges of the matching. Since a perfect matching has size $|V(G)|/2$, then $\alpha(G) \leqslant |V(G)|/2$, establishing equality and the result.

($\Rightarrow$) If $\alpha(G) = |V(G)|/2$, then U and W must have the same size. Let X be an arbitrary subset of U. By definition, $X \cup (W\backslash N(X))$ is an independent set, and the union is clearly disjoint. Therefore,

$$|V(G)|/2 = \alpha(G) \geqslant |X \cup (W\backslash N(X))| = |X| + |W\backslash N(X)| = |X| + |V(G)|/2 - |N(X)|$$

Thus, $|N(X)| \geqslant |X|$ and since X was arbitrary, Hall's theorem gives a matching that saturates U, hence of size $|U| = |V(G)|/2$, and therefore perfect. □